

a) Check that $u = \frac{1}{16\pi^4 - 4\pi^2} \sin(2\pi x)$ solves the following eq. on $[0, 1]$,

$$d_x^4 u + d_x^2 u = \sin(2\pi x)$$

where $u(0) = u(1) = 0$.

To check this we need to verify two things:

1. u satisfies the differential eq.

$$d_x u = \frac{2\pi}{16\pi^4 - 4\pi^2} \cos(2\pi x) = \frac{1}{8\pi^3 - 2\pi} \cos(2\pi x)$$

$$d_x^2 u = -\frac{2\pi}{8\pi^3 - 2\pi} \sin(2\pi x) = -\frac{1}{4\pi^2 - 1} \sin(2\pi x)$$

$$d_x^3 u = -\frac{2\pi}{4\pi^2 - 1} \cos(2\pi x)$$

$$d_x^4 u = \frac{4\pi^2}{4\pi^2 - 1} \sin(2\pi x)$$

$$\begin{aligned} d_x^4 u - d_x^2 u &= \frac{4\pi^2}{4\pi^2 - 1} \sin(2\pi x) - \frac{1}{4\pi^2 - 1} \sin(2\pi x) \\ &= \frac{4\pi^2 - 1}{4\pi^2 - 1} \sin(2\pi x) = \underline{\sin(2\pi x)} \Rightarrow \text{OK!} \end{aligned}$$

2. u satisfies the boundary conditions $u(0) = u(1) = 0$

$$u(0) = \frac{1}{16\pi^4 - 4\pi^2} \sin(0) = 0$$

$$u(1) = \frac{1}{16\pi^4 - 4\pi^2} \sin(2\pi) = 0 \Rightarrow \text{OK!}$$

Hence we can conclude that u solves the given diff. eq.

b) Determine the constant $\alpha \in \mathbb{R}$ so that $u = \frac{1}{16\pi^4 - 4\pi^2} \sin(2\pi x) + \alpha \sin(4\pi x)$ solves

$$d_x^4 u + d_x^2 u = \sin(2\pi x) + \sin(4\pi x), \quad u(0) = u(1) = 0.$$

$$d_x u = \frac{1}{8\pi^3 - 2\pi} \cos(2\pi x) + 4\pi\alpha \cos(4\pi x)$$

$$d_x^2 u = -\frac{1}{4\pi^2 - 1} \sin(2\pi x) - 16\pi^2\alpha \sin(4\pi x)$$

$$d_x^3 u = -\frac{2\pi}{4\pi^2 - 1} \cos(2\pi x) - 64\pi^3\alpha \cos(4\pi x)$$

$$d_x^4 u = \frac{4\pi^2}{4\pi^2 - 1} \sin(2\pi x) + 256\pi^4\alpha \sin(4\pi x)$$

$$d_x^4 u - d_x^2 u = \sin(2\pi x) + 256\pi^4\alpha \sin(4\pi x) - 16\pi^2\alpha \sin(4\pi x)$$

$$= \sin(2\pi x) + \alpha \sin(4\pi x) (256\pi^4 - 16\pi^2)$$

$$= \sin(2\pi x) + \sin(4\pi x), \quad \alpha = \frac{1}{256\pi^4 - 16\pi^2}$$

In this problem we will consider the heat eq. on $[0,1]$:

$$d_t u - d_x^2 u = f(x). \quad (1)$$

a) Consider (1) when $f(x) = 0$. Show that,

$$u_1 = e^{-4\pi^2 t} \cos(2\pi x), \quad \text{and} \quad u_2 = e^{-16\pi^2 t} \cos(4\pi x)$$

solves (1) in this case.

checks if u_1 satisfies the differential eq

$$d_t u_1 = -4\pi^2 e^{-4\pi^2 t} \cos(2\pi x)$$

$$d_x u_1 = -2\pi e^{-4\pi^2 t} \sin(2\pi x)$$

$$d_x^2 u_1 = -4\pi^2 e^{-4\pi^2 t} \cos(2\pi x)$$

$$\begin{aligned} d_t u_1 - d_x^2 u_1 &= -4\pi^2 e^{-4\pi^2 t} \cos(2\pi x) - (-4\pi^2 e^{-4\pi^2 t} \cos(2\pi x)) \\ &= 0 = \underline{f(x)} \Rightarrow \text{OK!} \end{aligned}$$

$$d_t u_2 = -16\pi^2 e^{-16\pi^2 t} \cos(4\pi x)$$

$$d_x u_2 = -4\pi e^{-16\pi^2 t} \sin(4\pi x)$$

$$d_x^2 u_2 = -16\pi^2 e^{-16\pi^2 t} \cos(4\pi x)$$

$$\begin{aligned} d_t u_2 - d_x^2 u_2 &= -16\pi^2 e^{-16\pi^2 t} \cos(4\pi x) - (-16\pi^2 e^{-16\pi^2 t} \cos(4\pi x)) \\ &= 0 = \underline{f(x)} \Rightarrow \text{OK!} \end{aligned}$$

b) Argue that $u_1 + u_2$ is also a solution to (1) in this case.

Since both u_1 and u_2 solves (1) we can argue that $u_1 + u_2$ is a solution as well because of superposition.

c) Set $f(x) = \sin(2\pi x)$. Show that

$$u_3 = \frac{1}{4\pi^2} (1 - e^{-4\pi^2 t}) \sin(2\pi x) = \frac{1}{4\pi^2} (\sin(2\pi x) - e^{-4\pi^2 t} \sin(2\pi x))$$

is a solution to (1) in this case. Is $u_3 + u_3$ a solution to (1) in this case?

checks if u_3 satisfies the differential eq

$$d_t u_3 = \frac{1}{4\pi^2} (0 + 4\pi^2 e^{-4\pi^2 t} \sin(2\pi x)) = e^{-4\pi^2 t} \sin(2\pi x)$$

$$d_x u_3 = \frac{1}{4\pi^2} (2\pi \cos(2\pi x) - 2\pi e^{-4\pi^2 t} \cos(2\pi x))$$

$$d_x^2 u_3 = \frac{1}{4\pi^2} (-4\pi^2 \sin(2\pi x) + 4\pi^2 e^{-4\pi^2 t} \sin(2\pi x)) = \sin(2\pi x) (e^{-4\pi^2 t} - 1)$$

$$d_t u_3 - d_x^2 u_3 = e^{-4\pi^2 t} \sin(2\pi x) - \sin(2\pi x) (e^{-4\pi^2 t} - 1)$$

$$= \sin(2\pi x) = \underline{f(x)} \Rightarrow \text{OK!}$$

Because of superpos., $u + u$ will also be a solution.

a) We now impose boundary conditions and an initial condition on (1):

$$u(0, x) = g(x), \quad u(t, 0) = u(t, 1) = 0.$$

Set $f(x) = \sin(2\pi x)$ and $g(x) = 0$, and check that u_3 solves (1) with these boundary conditions, and the initial condition.

To check this we need to verify two things:

1. u satisfies the differential eq.

2. u satisfies the boundary conditions $u(0) = u(1) = 0$

We confirmed 1. in c). Then we just have to check 2.

$$u_3(t, x) = \frac{1}{4\pi^2} (1 - e^{-4\pi^2 t}) \sin(2\pi x)$$

$$u_3(0, x) = \frac{1}{4\pi^2} (1 - 1) \sin(2\pi x) = 0 = \underline{g(x)} \Rightarrow \text{initial condition OK!}$$

$$u_3(t, 0) = \frac{1}{4\pi^2} (1 - 1) \sin(0) = \underline{0}$$

$$u_3(t, 1) = \frac{1}{4\pi^2} (1 - 1) \sin(2\pi) = \underline{0} \Rightarrow \text{boundary conditions OK!}$$

Hence we can conclude that u_3 solves the PDE.

e) Now, set $f(x) = 0$ and $g(x) = \sin(4\pi x)$, and denote the solution to (1) by u_4 . Check

$$u_4 = e^{-16\pi^2 t} \sin(4\pi x)$$

As in the previous tasks we have to check that u satisfies the two conditions mentioned earlier

$$\partial_t u_3 = -16 e^{-16\pi^2 t} \sin(4\pi x)$$

$$\partial_x u_3 = 4\pi e^{-16\pi^2 t} \cos(4\pi x)$$

$$\partial_x^2 u_3 = -16 e^{-16\pi^2 t} \sin(4\pi x)$$

$$\partial_t u_3 - \partial_x^2 u_3 = -16 e^{-16\pi^2 t} \sin(4\pi x) - (-16 e^{-16\pi^2 t} \sin(4\pi x)) = 0 = \underline{f(x)} \Rightarrow \text{OK!}$$

$$u_4(0, x) = 1 \cdot \sin(4\pi x) = \underline{g(x)} \Rightarrow \text{initial condition OK!}$$

$$u_4(t, 0) = e^{-16\pi^2 t} \sin(0) = \underline{0}$$

$$u_4(t, 1) = e^{-16\pi^2 t} \sin(4\pi) = \underline{0} \Rightarrow \text{boundary conditions OK!}$$

Hence we can conclude that u_4 solves the PDE.

Prob. 3 - Classification of PDE's

Classify the following PDE's into linear/nonlinear, homogeneous/inhomogeneous.

Linear: only linear operations and degree of one

Homogeneous: all terms have the same total degree and no constant term

a) One dimensional heat eq.

$$\partial_t u = \partial_x^2 u$$

$\Rightarrow$ Linear and homogeneous when right equals zero, else inhomogeneous

b) One dimensional heat eq. on $[0, 1]$ with varying diffusion

$$\partial_t u = x(x-1) \partial_x^2 u$$

$\Rightarrow$ Linear and homogeneous

c) The Laplace eq in polar coordinates ($u = u(r, \theta)$)

$$\partial_r(r \partial_r u) + \frac{1}{r} \partial_\theta^2 u = 0$$

$\Rightarrow$ Linear and homogeneous

d) The Poisson eq describing torsion of prismatic bars

$$\partial_x^2 u + \partial_y^2 u + 3 = 0$$

⇒ linear and inhomogeneous

e) The convection - reaction eq., with a non-zero reaction exponent $n \in \mathbb{N}$

$$\partial_t u + \partial_x u + u^n = 0$$

⇒ nonlinear and inhomogeneous

f) The bi-harmonic eq. in two dimensions

$$\partial_x^4 u + 8 \partial_x^2 \partial_y^2 u + \partial_y^4 u = 0$$

⇒ linear and homogeneous

Prob 4 - Wave eq.

$$\partial_t^2 u = \frac{FL}{m} \partial_x^2 u$$

$$u(x, 0) = \left(\frac{1}{2} + \left\| \frac{2x}{L} - 1 \right\| - 1 \right) - \left\| \frac{2x}{L} - 1 \right\| - \frac{1}{2} \Big) H.$$

a) Determine the wave speed c in terms of the given constants F , L and m .

By the def of the wave eq. we have

$$\partial_t^2 u = c^2 \partial_x^2 u \Rightarrow \partial_t^2 u = \frac{FL}{m} \partial_x^2 u \Rightarrow c = \sqrt{\frac{FL}{m}}$$

b) Determine the vibration period in terms of F , m and L .

To find the vibration period, we need to know the wavespeed and length.

$$\overset{\text{wavespeed}}{c} = \sqrt{\frac{FL}{m}} \quad \overset{\text{wavelength}}{\lambda} = 2L$$

Then we use this formula to calculate the vibration period

$$T = \frac{\lambda}{v} = \underline{2\sqrt{\frac{Lm}{F}}}$$